

Genome Sequencing is a hard problem

DNA is double stranded

Machines can error

Potentially multiple chromosomes

Imperfect "coverage"

Ignore all those errors for now.

Input: A collection of strings patterns each with length k k-mer is a k length substring

Output: A string Text whose k-mer composition is equal to patterns

General strategies:

Check prefixes and suffixes.

We need them to match. If the prefix/suffix doesn't match then we know that it starts/ends w/ that string.

Half of human genome is repeats.

We can view matching prefix/suffix as a directed edge in a graph.

Task boils down to finding a Hamiltonian Path in this larger graph.

However, this is NP-Complete.